

NORYL™ Resin GFN1F - Asia

Polyphenylene Ether + PS

SABIC

PROSPECTOR®

www.ulprospector.com

Technical Data

Product Description

NORYL GFN1F resin is a 10% glass fiber reinforced blend of polyphenylene ether (PPE) + polystyrene (PS). This injection moldable grade exhibits very low moisture absorption, high strength, hydrolytic stability, Low warpage, low specific gravity, and dimensional stability. NORYL GFN1F carries a UL746C outdoor suitability rating of F1 along with FDA food contact compliance and NSF 61 listings in several colors. The properties of this material makes it an excellent candidate for water management applications such as water filter and meter components, pump housings / impellers, shower + faucet, and valves.*See NORYL GFN1 resin for NON FDA / NSF version.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E45587-237019 • E207780-228544		
Search for UL Yellow Card	• SABIC • NORYL™ Resin		
Availability	• Asia Pacific		
Filler / Reinforcement	• Glass Fiber, 10% Filler by Weight		
Features	• Amorphous • Corrosion Resistant • Food Contact Acceptable • Good Dimensional Stability	• Good Stiffness • Hydrolytically Stable • Low Moisture Absorption • Low Shrinkage	• Low Specific Gravity • Low Warpage
Uses	• Hygiene • Personal Care	• Potable Water Applications • Water Management	
Processing Method	• Injection Molding		
Also Available In	• Latin America	• North America	

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity	1.17 g/cm³	ASTM D792 ISO 1183
Melt Mass-Flow Rate (MFR) (280°C/5.0 kg)	4.0 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (280°C/5.0 kg)	4.0 cm³/10min	ISO 1133
Molding Shrinkage - Flow (3.20 mm)	0.55 to 0.66 %	Internal Method
Water Absorption		ISO 62
Saturation, 23°C	0.12 %	
Equilibrium, 23°C, 50% RH	0.040 %	
Outdoor Suitability	f1	UL 746C

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus		
-- ³	4400 MPa	ASTM D638
--	3900 MPa	ISO 527-1/1
Tensile Strength		
Yield ⁴	75.0 MPa	ASTM D638
Yield	60.0 MPa	ISO 527-2/50
Break ⁴	75.0 MPa	ASTM D638
Break	60.0 MPa	ISO 527-2/50
Tensile Elongation		
Yield ⁴	3.5 %	ASTM D638
Yield	4.5 %	ISO 527-2/50
Break ⁴	3.5 %	ASTM D638
Break	4.5 %	ISO 527-2/50
Flexural Modulus		
50.0 mm Span ⁵	3400 MPa	ASTM D790
-- ⁶	3700 MPa	ISO 178



NORYL™ Resin GFN1F - Asia

Polyphenylene Ether + PS

SABIC**PROSPECTOR®**

www.ulprospector.com

Mechanical	Nominal Value Unit	Test Method
Flexural Stress		
-- 6, 7	115 MPa	ISO 178
Yield, 50.0 mm Span ⁵	105 MPa	ASTM D790
Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength ⁸ (23°C)	8.0 kJ/m ²	ISO 179/1eA
Notched Izod Impact		
-30°C	80 J/m	ASTM D256
23°C	84 J/m	ASTM D256
-30°C ⁹	9.0 kJ/m ²	ISO 180/1A
23°C ⁹	10 kJ/m ²	ISO 180/1A
Unnotched Izod Impact		
23°C	410 J/m	ASTM D4812
-30°C ⁹	30 kJ/m ²	ISO 180/1U
23°C ⁹	30 kJ/m ²	ISO 180/1U
Instrumented Dart Impact		
23°C, Total Energy	20.0 J	ASTM D3763
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
0.45 MPa, Unannealed, 3.20 mm	135 °C	ASTM D648
1.8 MPa, Unannealed, 3.20 mm	130 °C	ASTM D648
1.8 MPa, Unannealed, 4.00 mm, 64.0 mm Span ⁹	129 °C	ISO 75-2/Af
Vicat Softening Temperature		
--	138 °C	ASTM D1525 ¹⁰ ISO 306/B50 ¹⁰
--	140 °C	ISO 306/B120
CLTE		
Flow : -40 to 40°C	4.5E-5 cm/cm/°C	ASTM E831
Flow : -40 to 40°C	3.8E-5 cm/cm/°C	ISO 11359-2
Transverse : -40 to 40°C	8.2E-5 cm/cm/°C	ASTM E831
Transverse : -40 to 40°C	8.5E-5 cm/cm/°C	ISO 11359-2
RTI Elec	90.0 °C	UL 746B
RTI Imp	90.0 °C	UL 746B
RTI Str	90.0 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Arc Resistance ¹¹	PLC 6	ASTM D495
Comparative Tracking Index (CTI)	PLC 3	UL 746A
High Amp Arc Ignition (HAI)		UL 746A
> 1.5 mm	PLC 3	
> 3.0 mm	PLC 4	
High Voltage Arc Resistance to Ignition (HVAR)	PLC 4	UL 746A
Hot-wire Ignition (HWI)		UL 746A
> 1.5 mm	PLC 1	
> 3.0 mm	PLC 2	
Flammability	Nominal Value Unit	Test Method
Flame Rating (> 1.5 mm)	HB	UL 94
Injection	Nominal Value Unit	
Drying Temperature	105 to 110 °C	
Drying Time	3.0 to 4.0 hr	
Suggested Max Moisture	0.020 %	
Suggested Shot Size	30 to 70 %	
Rear Temperature	260 to 305 °C	



NORYL™ Resin GFN1F - Asia

Polyphenylene Ether + PS

SABIC

PROSPECTOR®

www.ulprospector.com

Injection	Nominal Value Unit
Middle Temperature	270 to 310 °C
Front Temperature	280 to 315 °C
Nozzle Temperature	295 to 315 °C
Processing (Melt) Temp	295 to 315 °C
Mold Temperature	75 to 105 °C
Back Pressure	0.300 to 0.700 MPa
Screw Speed	20 to 100 rpm

Injection Notes

- Drying Time (Cumulative): 8 hr

Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ 5.0 mm/min

⁴ Type I, 50 mm/min

⁵ 1.3 mm/min

⁶ 2.0 mm/min

⁷ at Yield

⁸ 80*10*4 sp=62mm

⁹ 80*10*4 mm

¹⁰ Rate A (50°C/h), Loading 2 (50 N)

¹¹ Tungsten Electrode



NORYL™ Resin GFN1F - Asia

Polyphenylene Ether + PS

SABIC

PROSPECTOR®

www.ulprospector.com

Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

3Polymer (Guangzhou) Chemical Technology Co., Ltd.

Telephone: +86-20-3466-7988

Web: <http://3polymer.com>

Availability: China

